PERFORMANCEINTELLIGENCE

ForecastIQ

Probabilistic revenue planning and risk-aware budget guidance — 30, 60, and 90-day outlook

AUDITDATE	DATAQUALITY	RISKCLASSIFICATION	PLATFORMMODE
2026-07-10	**100.0/100**	**High Risk**	**Offline Utility**

1 Executive Summary

KEYTAKEAWAY

Over the full historical timeline, ForecastIQ has audited $2.18M in digital marketing spend yielding $11.10M in revenue (Portfolio ROAS: 5.09x). Our 90-day probabilistic ensemble projects sustained top-line stability with a P50 expected capture of $0.84M. Search campaigns continue to function as your high-efficiency anchor, while social prospecting provides scalable incremental demand.

2 Multi-Horizon Revenue & ROAS Projections

The following forecasts are derived from a weighted ensemble (Prophet, XGBoost, LightGBM, CatBoost). P10 is a conservative floor, P50 the expected baseline, and P90 an upside scenario — not point estimates, but a probability range meant to be planned against.

Planning Window	Metric	P10 (Conservative)	P50 (Expected)	P90 (Upside)
30 Days	**Revenue**	$171,113	**$278,581**	$386,049
30 Days	**ROAS**	2.94x	**4.79x**	6.64x
60 Days	**Revenue**	$380,070	**$560,810**	$741,549
60 Days	**ROAS**	3.23x	**4.77x**	6.31x
90 Days	**Revenue**	$599,967	**$844,942**	$1,089,916
90 Days	**ROAS**	3.40x	**4.78x**	6.17x

3 Forecast Accuracy: Offline Backtest vs. Naive Baseline

3-fold rolling-origin

To check whether the uncertainty bands above are meaningful rather than arbitrary, the ensemble was evaluated with a rolling-origin backtest: retrained on progressively larger historical windows and scored against forecast periods it had not seen, then compared to a naive baseline (a flat projection of the trailing 30-day daily average). Figures below are computed at the Overall (portfolio) level, where percentage-based accuracy is stable; Campaign-level rows are noisier and are addressed separately below.

Window	Metric	Model MAPE	Naive Baseline MAPE	P10–P90 Coverage	Result
30 Days	**ROAS**	24.1%	23.8%	67%	**Trails baseline** (-1.1%)
30 Days	**Revenue**	21.3%	17.6%	33%	**Trails baseline** (-20.9%)
60 Days	**ROAS**	12.7%	13.3%	67%	**Beats baseline** (+4.5%)
60 Days	**Revenue**	20.5%	18.4%	33%	**Trails baseline** (-11.5%)
90 Days	**ROAS**	19.3%	18.0%	67%	**Trails baseline** (-7.1%)
90 Days	**Revenue**	8.8%	6.6%	67%	**Trails baseline** (-33.2%)

ROAS forecasts beat the naive baseline at the 60-day horizon (+4.5% MAPE improvement) and trail it narrowly at 30 and 90 days (-1.1% and -7.1%). Revenue forecasts trail the naive baseline across all three horizons in this backtest. We're disclosing this rather than omitting it. Tracing the gap by decomposing the forecast into its components found that the model's own recency-anchoring step — which blends the raw ensemble output with a trailing revenue baseline to stabilize it — was weighting a stale 90-day average too heavily relative to the more recent 30-day level the naive baseline itself uses, overshooting by 18–23% APE on its own before the underlying model even contributed. Re-weighting that anchor toward the same recency the naive baseline uses cut the Revenue MAPE gap by roughly 60–75% across all three horizons. Two further factors widen the remaining gap: (1) the earliest backtest origin trains on only ≈ 4 months of history — well short of what Prophet's seasonal component needs to calibrate reliably, and (2) the dataset contains a genuine 5–10x November–December demand spike that appears in only two historical instances across the ≈ 2.5-year dataset, so any forecast horizon crossing that window is working with minimal precedent — a constraint that affects any forecasting method, not just this one.

KNOWNLIMITATIONS

Forecast accuracy improves with training history and is weakest for the earliest, data-sparse backtest origin. November–December forecasts carry wider uncertainty given only two observed instances of the holiday demand spike in the historical record. Campaign-level forecasts (excluded from the table above) inherit materially higher variance than portfolio-level ones, since individual campaign lifecycles and end-dates aren't modeled as an explicit signal.

4 Channel Performance & Contribution Breakdown

Treating existing channel attribution as the source of truth, here is the forward 60-day expected contribution and efficiency analysis across your primary paid acquisition channels.

Acquisition Channel	Metric	P10	P50 (Expected)	P90
Bing Ads	Revenue	$546	**$10,925**	$28,916
Bing Ads	ROAS	0.08x	**1.63x**	4.31x
Google Ads	Revenue	$162,504	**$516,646**	$870,787
Google Ads	ROAS	1.62x	**5.14x**	8.66x
Meta Ads	Revenue	$3,723	**$74,450**	$232,668
Meta Ads	ROAS	0.26x	**5.21x**	16.28x

5 Budget Optimizer Recommendations

An Optuna-driven search allocated a monthly budget of **$100,000** to maximize total revenue subject to a target ROAS floor of **4.5x**. The search assumes diminishing marginal returns per channel — pushing more spend into one channel yields progressively less revenue per dollar — so the optimizer balances the split across all channels rather than concentrating on whichever looks best in isolation.

Channel	Recommended Spend	Expected Revenue	Expected ROAS	Spend Share
Bing Ads	$1,021	$4,860	**4.76x**	1.0%
Google Ads	$45,898	$230,191	**5.02x**	46.0%
Meta Ads	$52,897	$326,678	**6.18x**	53.0%
OPTIMIZED TOTAL	**$99,817**	**$561,730**	**5.63x**	**100.0%**

READINGTHISTABLE

If Meta spend rises 20%, revenue increases but marginal ROAS falls. The optimizer's job is to find the allocation that protects your ROAS floor while still increasing revenue — a business decision, not just a bigger number.

6 Risk Intelligence & Growth Action Plan

The overall portfolio risk classification is **High Risk** (risk score: 70.3/100). Below are the specific operational risks identified and recommended mitigation actions.

Risk Factor	Score	Status	Mitigation
Revenue Volatility	100.0/100	**Monitor**	Utilize automated pacing tools to smooth out mid-week revenue depressions and maintain consistent top-of-funnel impression share.
Channel Dependency	58.5/100	**Concentrated**	Google Ads represents 83.5% of revenue capture. Begin scaling top-of-funnel prospecting on your other channels to establish a strong secondary growth pillar.
ROAS Instability	85.4/100	**Volatile**	Enforce rigorous Target CPA / Target ROAS bid floor rules across all Meta generic and non-brand campaigns to prevent runaway spend.
Data Quality Audit	0.0/100	**Excellent**	Validation engine verified a 100.0/100 schema score. Ensure Bing campaign UTM tracking tags are consistently appended.

ForecastIQ — built for the NetElixir AIgnition 2026 Hackathon Challenge.

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